

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - Virtual Memory

hackerrank.com/test/g1lj816k4tp/questions/fdc576kjgji

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 21 / 36 01 hour 14 mins Rohan Dobarkar

2. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

- ☒ This memory address does not map to actual memory cell that corresponds to address number.
- ☐ This memory is the same as physical memory.
- ☐ This memory cannot be accessed by applications at run time.
- ☐ This memory address is always present even after a system is restarted.

Clear Selection

3. Consider the two functions incr and decr shown below. `incr(){ wait(s); wait(s); X = X+1; X = X-1; signal(s); signal(s); }`

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HackerRank Question - C x +

hackerrank.com/test/g1j816k4tp/questions/gdc5p221dlq

ORACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 22 / 36 01 hour 12 mins Rohan Dobarkar

3. Consider the two functions `incr` and `decr` shown below. `incr(){ decr(){ wait(s); wait(s); X = X+1; X = X-1; signal(s); signal(s); } }` There are 5 threads each invoking `incr` once, and 3 threads each invoking `decr` once, on the same shared variable `X`. The initial value of shared variable `X` is 10. Suppose there are two implementations of the semaphore `S` as follows: I-1: `S` is a binary semaphore initialized to 1. I-2: `S` is a counting semaphore initialized to 2. Let `V1` and `V2` be the values of `X` at the end of execution of all the threads with implementations I-1 and I2 respectively. Which one of the following choices corresponds to the minimum possible values of `V1` and `V2` respectively?

Consider the two functions `incr` and `decr` shown below.

```
incr(){
    wait(s);
    X = X+1;
    signal(s);
}

decr(){
    wait(s);
    X = X-1;
    signal(s);
}
```

There are 5 threads each invoking `incr` once, and 3 threads each invoking `decr` once, on the same shared variable `X`. The initial value of shared variable `X` is 10.

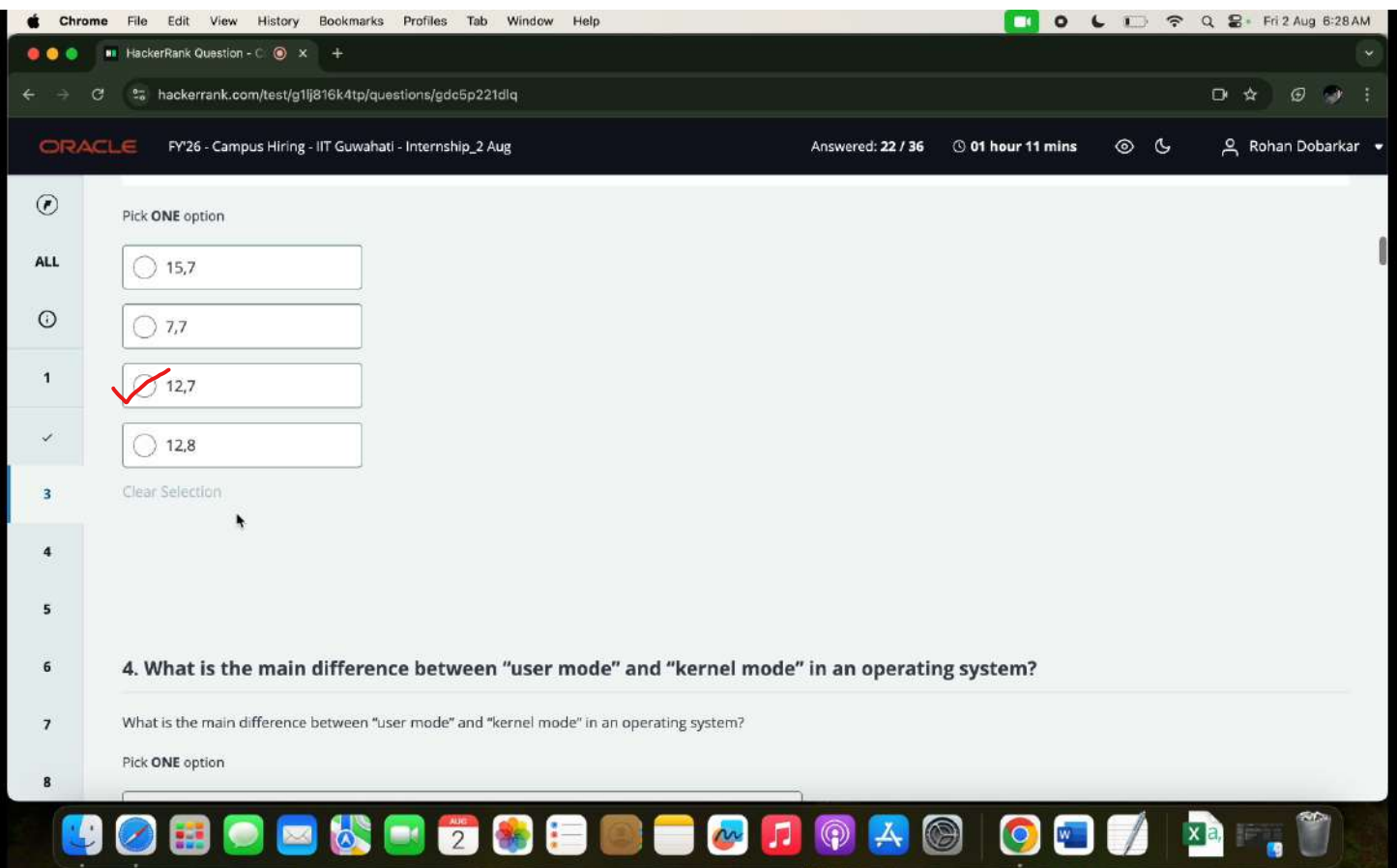
Suppose there are two implementations of the semaphore `S` as follows:

I-1: `S` is a binary semaphore initialized to 1.
I-2: `S` is a counting semaphore initialized to 2.

Let `V1` and `V2` be the values of `X` at the end of execution of all the threads with implementations I-1 and I2 respectively.

Which one of the following choices corresponds to the minimum possible values of `V1` and `V2` respectively?

https://www.hackerrank.com/test/g1j816k4tp/questions/gdc5p221dlq



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HackerRank Question - V x +

hackerrank.com/test/g1j816k4tp/questions/c3aaf70q0e6

ORACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 22 / 36 01 hour 11 mins Rohan Dobarkar

ALL

4. What is the main difference between "user mode" and "kernel mode" in an operating system?

What is the main difference between "user mode" and "kernel mode" in an operating system?

Pick **ONE** option

1 ☐ User mode allows direct hardware access, kernel mode does not.

✓ ☒ Kernel mode can access protected memory regions, user mode cannot.

3 ☐ Kernel mode applications are given less priority than user mode applications.

4 ☐ User mode instructions can switch the CPU to kernel mode, but not vice-versa.

5 Clear Selection

6

7

https://www.hackerrank.com/test/g1j816k4tp/questions/c3aaf70q0e6 What is suited for implementing a priority queue?

Calendar: AUG 2

Taskbar: Safari, Photos, Messages, Mail, App Store, Calendar, Photos, Music, Podcasts, App Store, System Settings, Chrome, Word, Notes, Excel, Trash

ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HackerRank Question - W

hackerrank.com/test/g1j816k4tp/questions/a7dn8s3ffm6

Fri 2 Aug 6:28 AM

ORACLEFY26 - Campus Hiring - IIT Guwahati - Internship_2 AugAnswered: 23 / 3601 hour 11 minsRohan Dobarkar

ALL

5

5. Which data structure is best suited for implementing a priority queue?

Which data structure is best suited for implementing a priority queue?

Pick ONE option

☐ Array

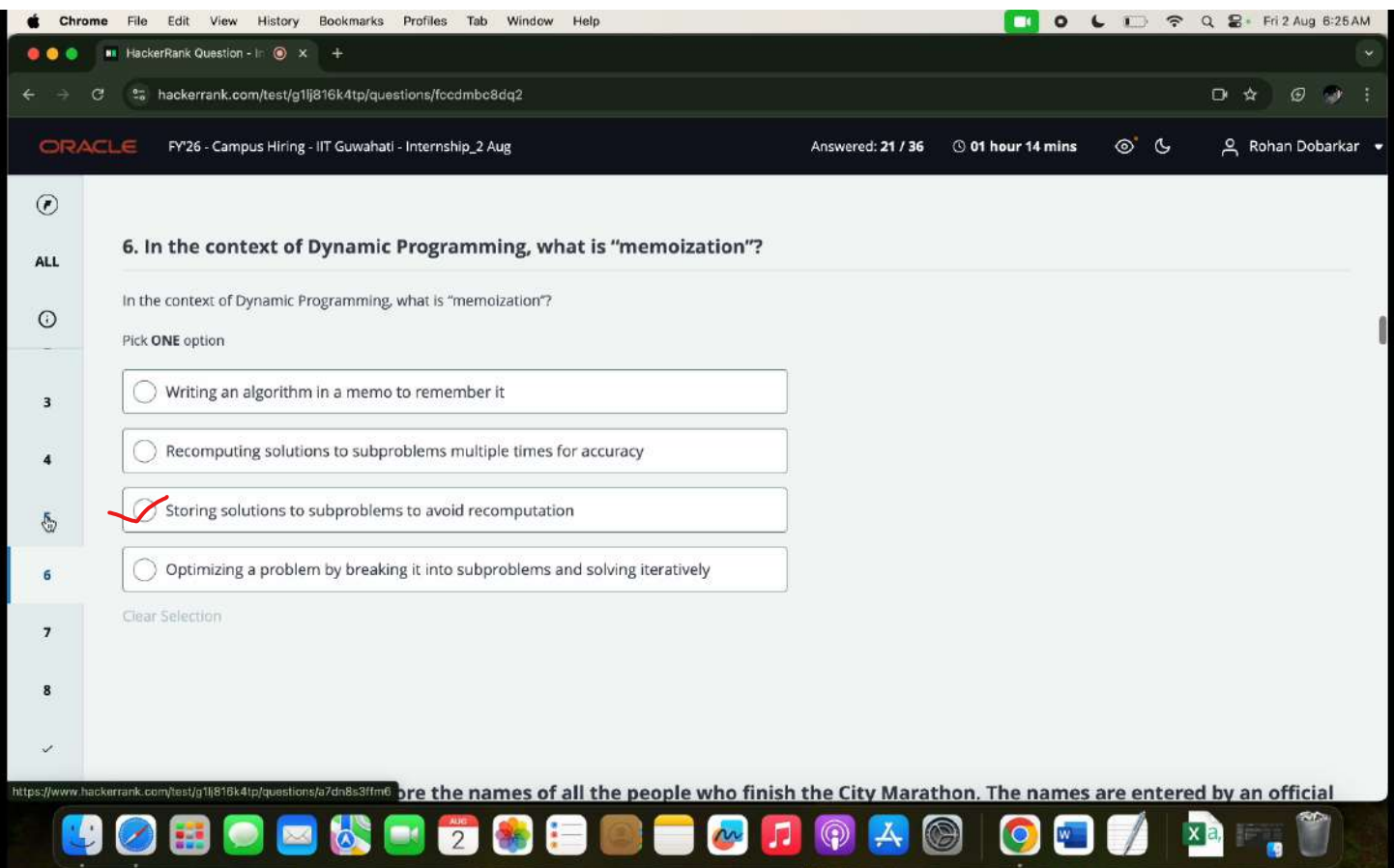
☐ Slack

☒ Heap

☐ Linked List

Clear Selection

programming, what is "memoization"?



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HackerRank Question - A x +

hackerrank.com/test/g1j816k4tp/questions/asnrkim1c2c

ORACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 21 / 36 01 hour 14 mins Rohan Dobarkar

7. A certain program must store the names of all the people who finish the City Marathon. The names are entered by an official at the finish line as the competitors finish. After the race is over the program must print a list only containing the names of competitors who placed 1 to 10 and then 100, 200, 300 and so on, as these competitors will receive a special medallion. Which data structure would be the best choice for this program?

A certain program must store the names of all the people who finish the City Marathon. The names are entered by an official at the finish line as the competitors finish. After the race is over the program must print a list only containing the names of competitors who placed 1 to 10 and then 100, 200, 300 and so on, as these competitors will receive a special medallion. Which data structure would be the best choice for this program?

Pick ONE option

☒ A vector or resizable array

☐ A map (implemented as a binary tree of key/value pairs)

☐ A singly linked list

☐ A hash set (implemented as a hash table of keys)

Clear Selection

https://www.hackerrank.com/test/g1j816k4tp/questions/ccdmibc8dq2

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - # x +

hackerrank.com/test/g1j816k4tp/questions/8n7mn8tdddmi

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 20 / 36 01 hour 15 mins Rohan Dobarkar

8. #include <stdio.h> int main() { char arr[] = "abcd"; char *p = arr; printf("%c\t", ++*p); printf("%c\t", *p++); printf("%c\t", (*p)++); printf("%c\n", *p); return 0; }

```
#include <stdio.h>
int main() {
    char arr[] = "abcd";
    char *p = arr;
    printf("%c\t", ++*p);
    printf("%c\t", *p++);
    printf("%c\t", (*p)++);
    printf("%c\n", *p);
    return 0;
}
```

8 Pick ONE option

☒ b b b c

☐ b c c d

☐ b b c c

https://www.hackerrank.com/test/g1j816k4tp/questions/asnikimfc2c

2

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 20 / 36 01 hour 15 mins Rohan Dobarkar

printf("%c\n", *p); return 0; }

```
#include <stdio.h>
int main() {
    char arr[] = "abcd";
    char *p = arr;
    printf("%c\t", *p);
    printf("%c\t", *p++);
    printf("%c\t", (*p)++);
    printf("%c\n", *p);
    return 0;
}
```

Pick **ONE** option

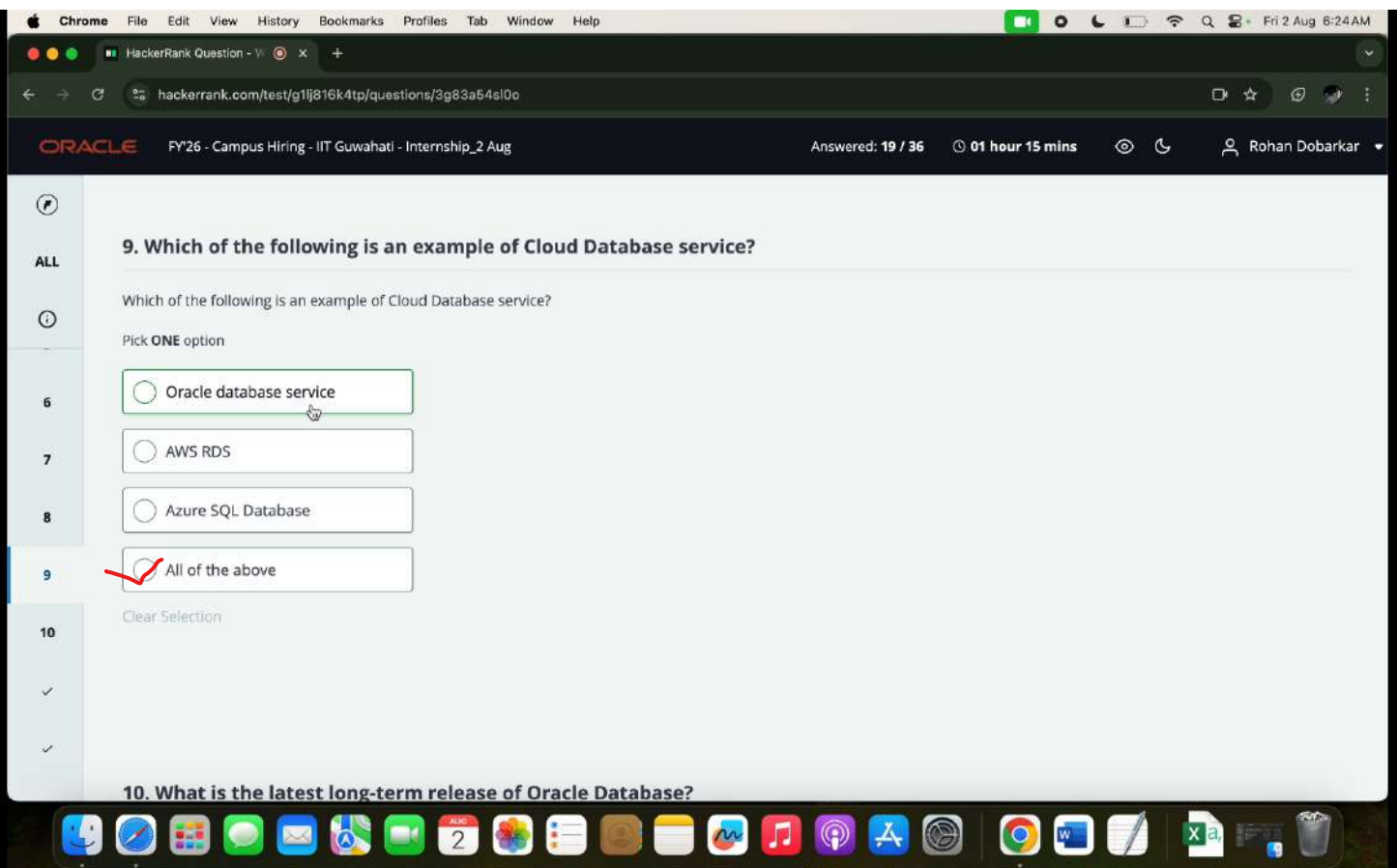
☐ b b b c

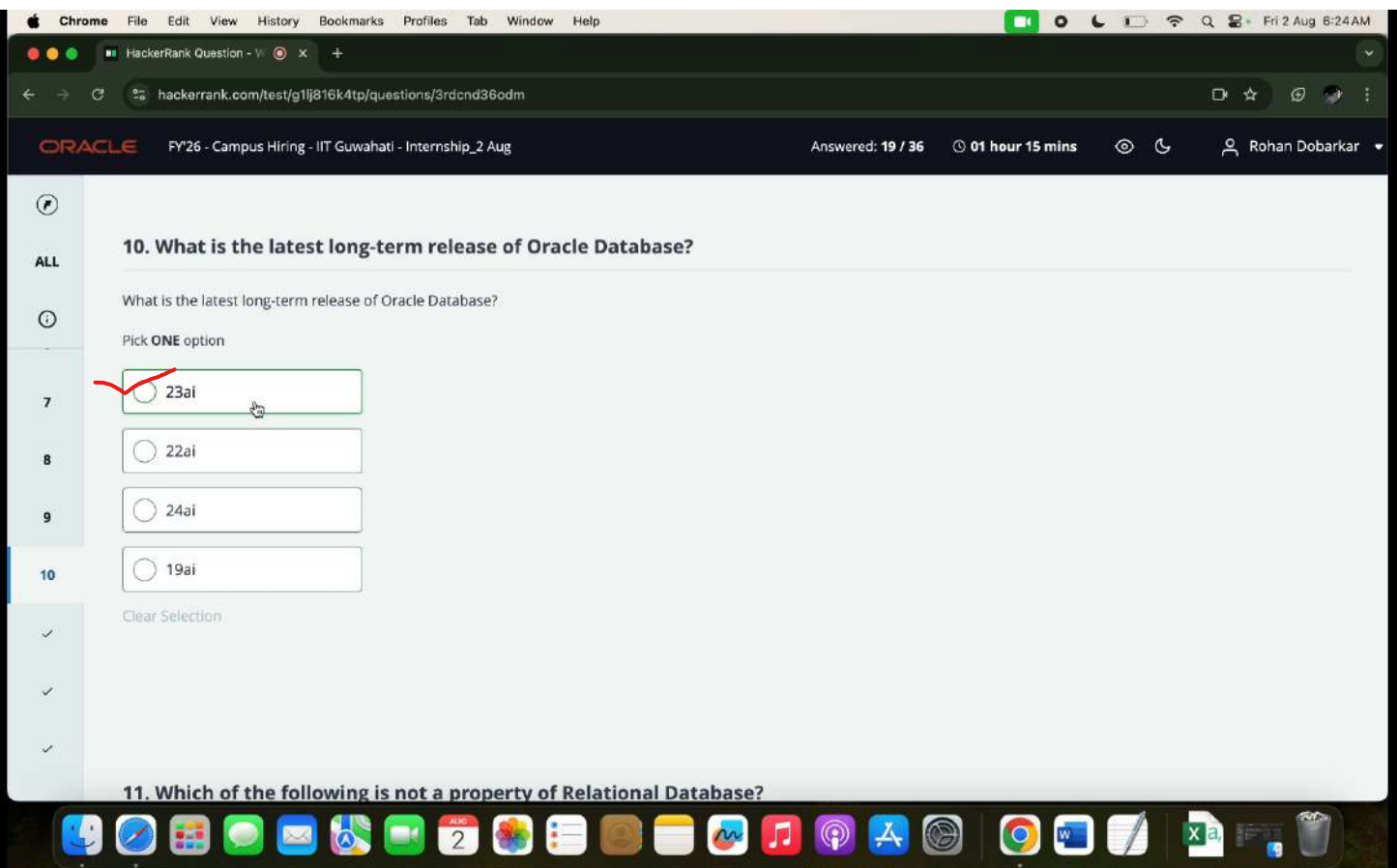
☐ b c c d

☐ b b c c

☐ b c c c

Clear Selection





Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - W x

hackerrank.com/test/g1lj816k4tp/questions/1enqc6edmc

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 18 / 36 01 hour 15 mins Rohan Dobarkar

ALL

11. Which of the following is not a property of Relational Database?

Which of the following is not a property of Relational Database?

Pick **ONE** option

8 ☐ Atomicity

9 ☐ Consistency

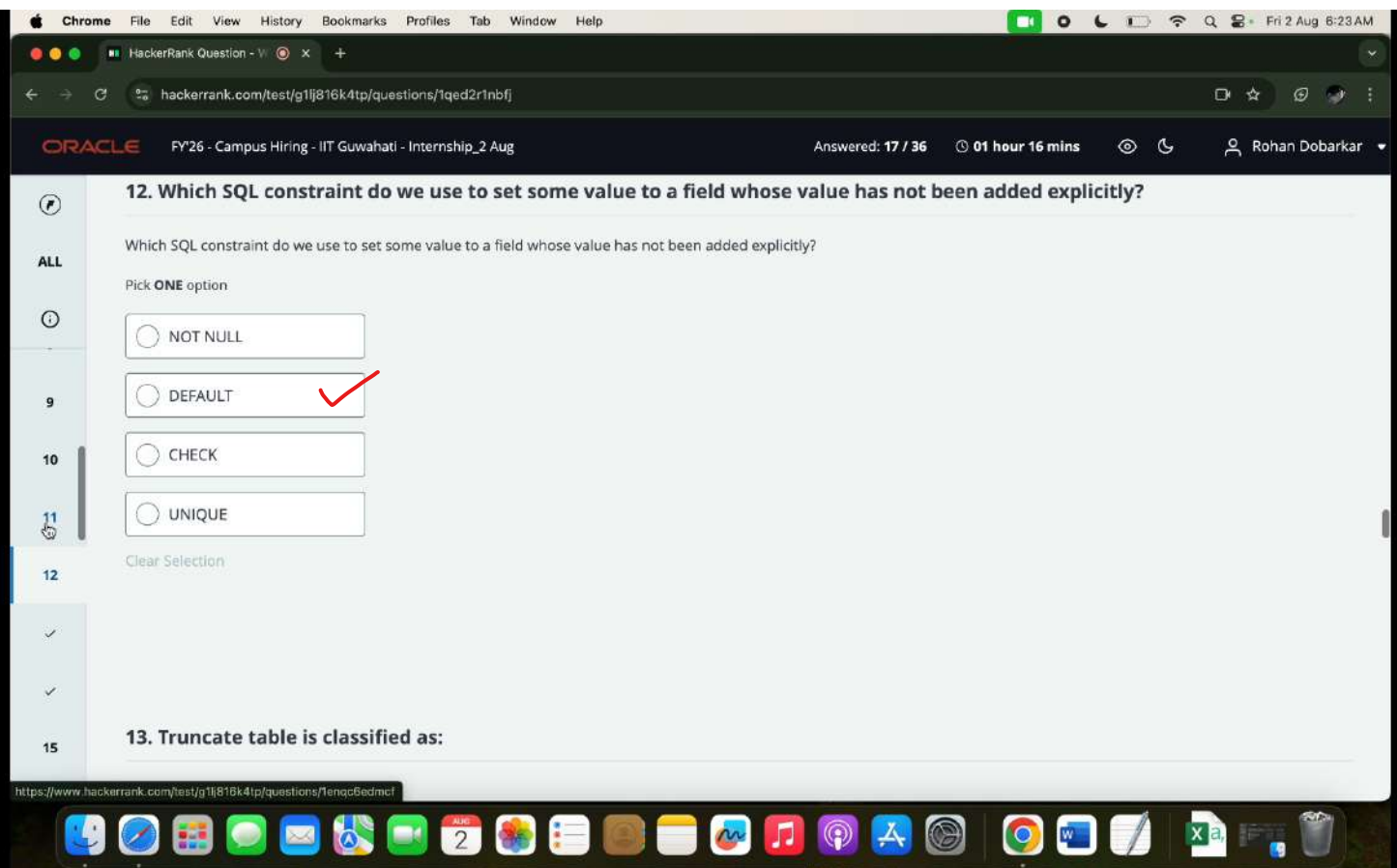
10 ☐ Isolation

11 ☒ Dependency

Clear Selection

12. Which SQL constraint do we use to set some value to a field whose value has not been added explicitly?

2



Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - Truncate table is classified as: x

hackerrank.com/test/g1lj816k4tp/questions/das6ocm4cld

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 16 / 36 01 hour 17 mins Rohan Dobarkar

ALL

13. Truncate table is classified as:

Truncate table is classified as:

Pick ONE option

10 ☒ Data Definition Language ✓

11 ☐ Data Manipulation Language ✗

12 ☐ Data Control Language

13 ☐ Transaction Control Language

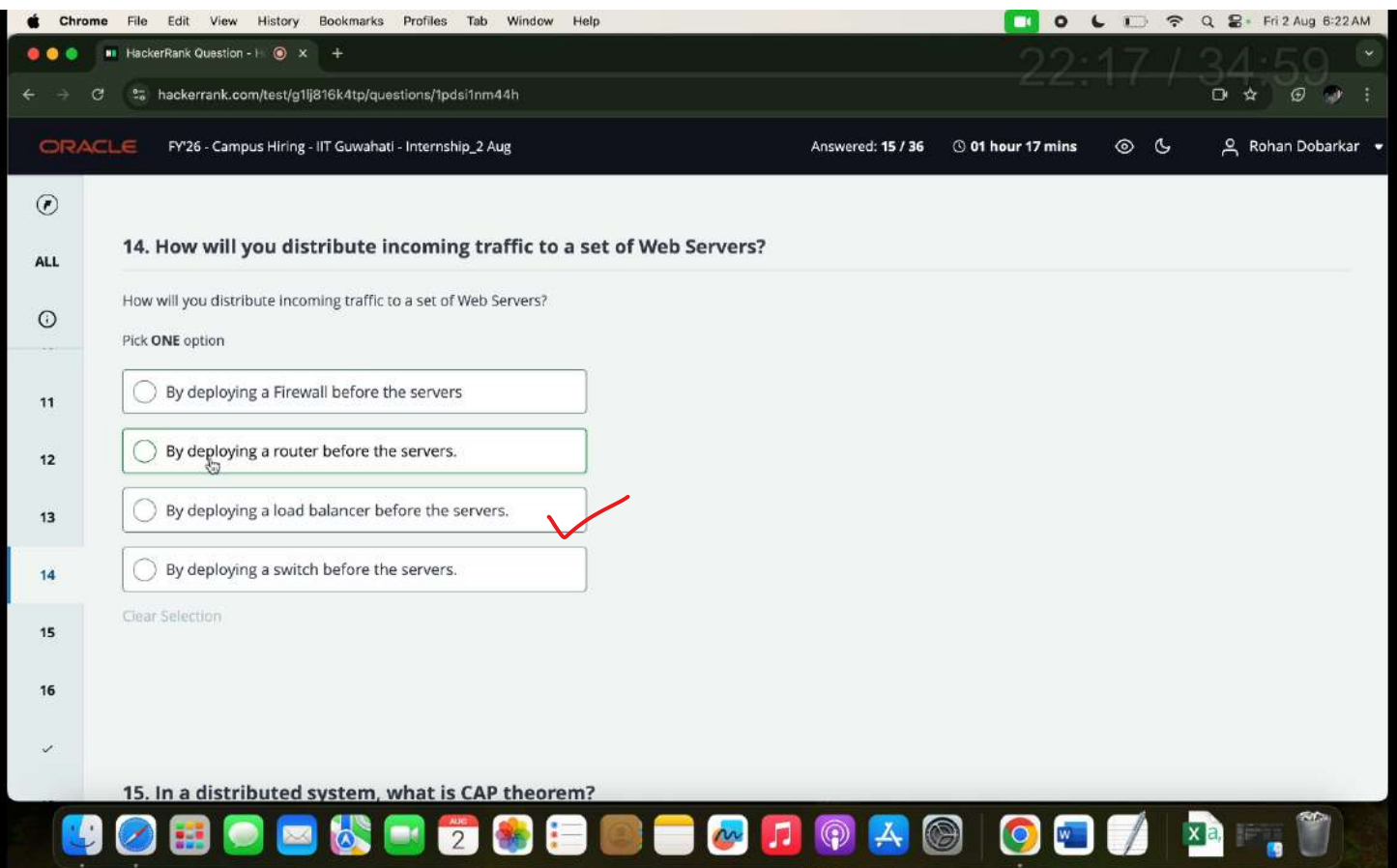
✓ Clear Selection

15

16

14. How will you distribute incoming traffic to a set of Web Servers?

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100



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HackerRank Question - In x +

hackerrank.com/test/g1j816k4tp/questions/3j9b31ro14

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 14 / 36 01 hour 17 mins Rohan Dobarkar

ALL

15. In a distributed system, what is CAP theorem?

In a distributed system, what is CAP theorem?

Pick **ONE** option

12 ☐ It specifies the maximum allowable response time for a system

13 ☒ It describes the trade-offs between consistency, availability, and partition tolerance. ✓

14 ☐ It defines the minimum number of servers required for fault tolerance.

15 ☐ It outlines the process for scaling a system horizontally.

Clear Selection

16

✓

18

16. What is a Zombie process in Linux ?

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

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HackerRank Question - W x +

hackerrank.com/test/g1lj816k4tp/questions/9pn6i7gft1k

21:57 / 34:59

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 13 / 36 01 hour 18 mins Rohan Dobarkar

16. What is a Zombie process in Linux ?

What is a Zombie process in Linux ?

Pick **ONE** option

13 ☐ A process which is running in the background.

14 ☐ A process which is running for a long time.

15 ☒ A process whose execution has completed but it still has an entry in the process table. ✓

16 ☐ A process which has crashed due to memory corruption.

Clear Selection

17. Which HTTP method is suitable for resource creation?

18

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - W x

hackerrank.com/test/g1lj816k4tp/questions/9jk9j5sdf4

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 12 / 36 01 hour 18 mins Rohan Dobarkar

ALL

17. Which HTTP method is suitable for resource creation?

Which HTTP method is suitable for resource creation?

Pick ONE option

14 ☐ PUT

15 ☐ PATCH

16 ☒ POST

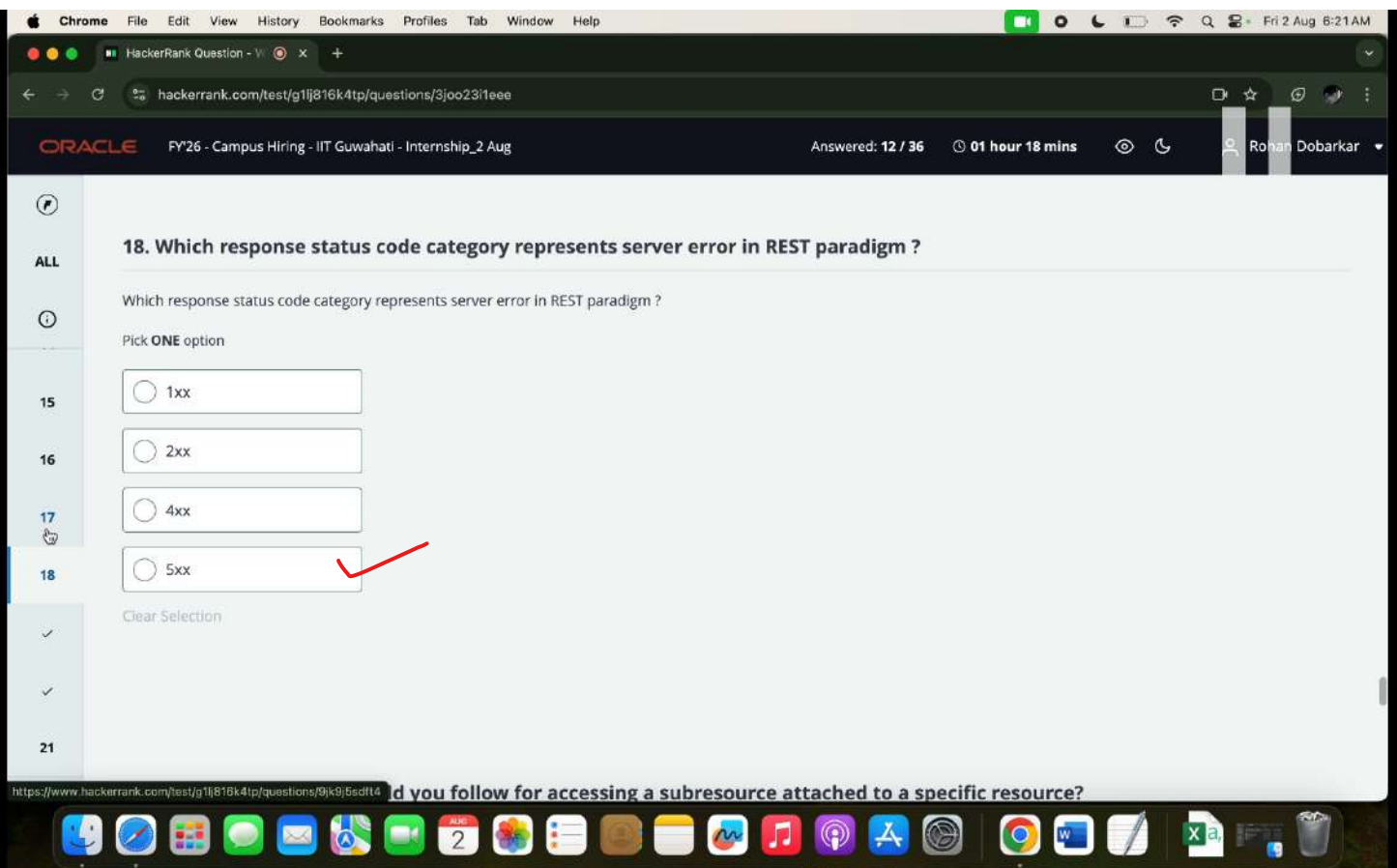
17 ☐ CREATE

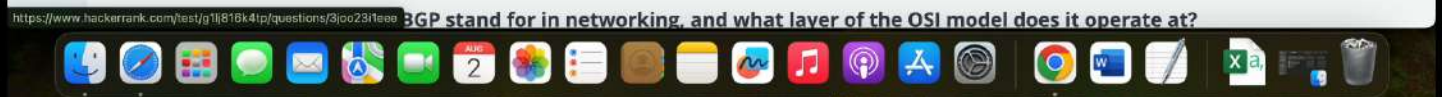
18 Clear Selection

18. Which response status code category represents server error in REST paradigm ?

2

Google Chrome, Microsoft Word, Microsoft Excel, and other application icons are visible in the dock.





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HackerRank Question - W x +

hackerrank.com/test/g1j816k4tp/questions/bj3hhe2fndn

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 10 / 36 01 hour 20 mins Ro Dobarkar

20. What does the acronym BGP stand for in networking, and what layer of the OSI model does it operate at?

What does the acronym BGP stand for in networking, and what layer of the OSI model does it operate at?

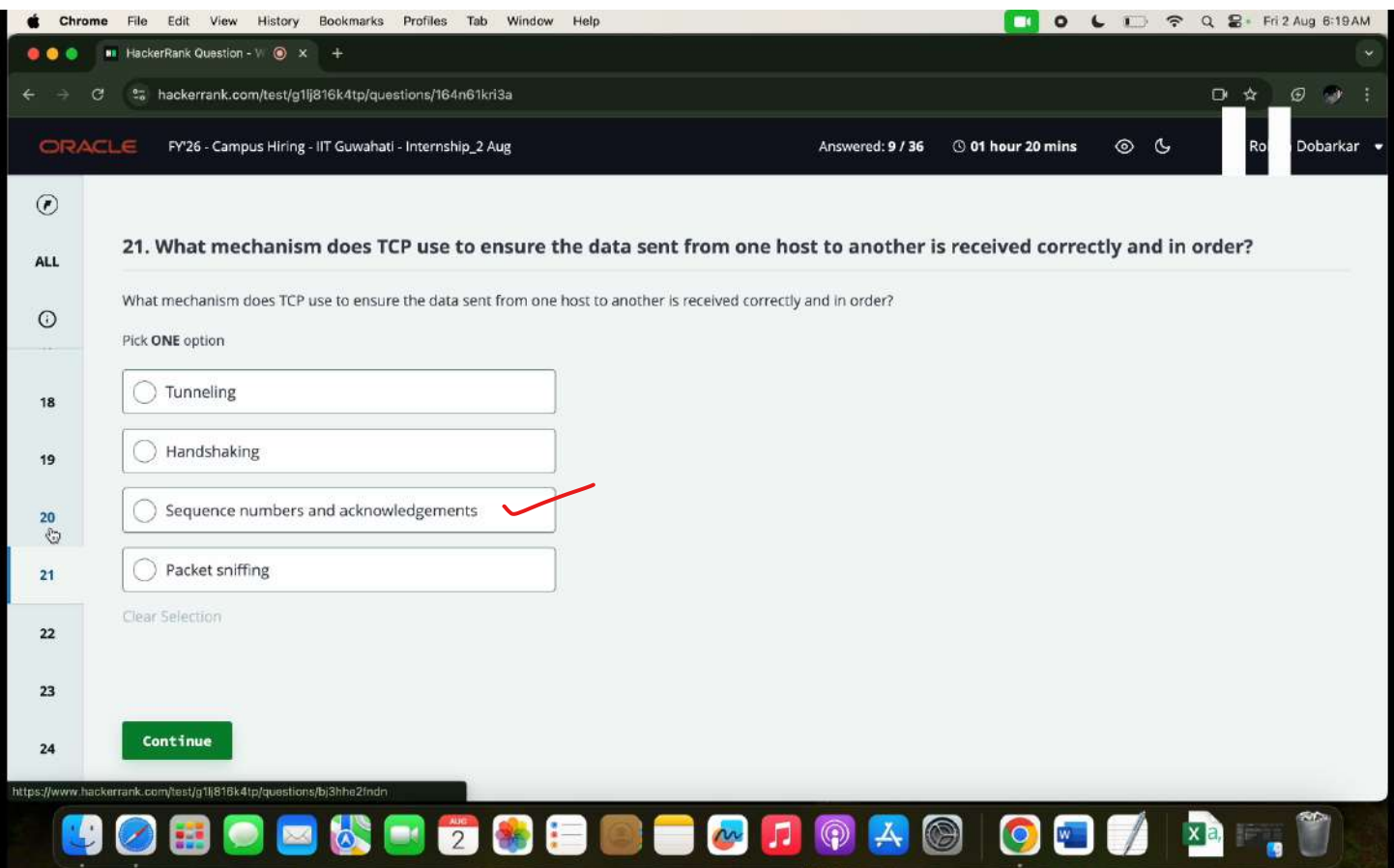
Pick **ONE** option

- ☐ Best Gateway Protocol, Network layer
- ☐ Border Gateway Protocol, Application layer
- ☒ Border Gateway Protocol, Network layer
- ☐ Backbone Gateway Protocol, Data Link layer

Clear Selection

https://www.hackerrank.com/test/g1j816k4tp/questions/3nqm1ek36n P use to ensure the data sent from one host to another is received correctly and in order?

2



ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HackerRank Question - Im x

hackerrank.com/test/g1lj816k4tp/questions/8aghtpsmcb

Fri 2 Aug 6:19 AM

ORACLE

FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 8 / 36

01 hour 20 mins

Rohan Dobarkar

ALL

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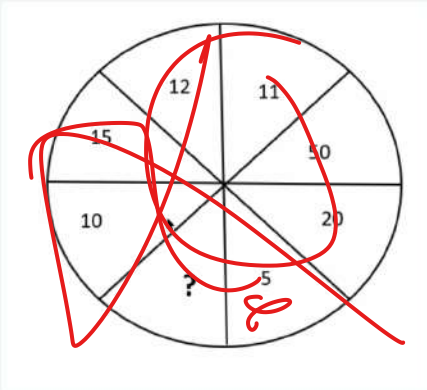
23

24

25

22. Improve the Sentence: My father is suffering from diabetes for the past three years.

Improve the Sentence: My father is suffering from diabetes for the past three years.



Pick ONE option

Waiting for www.hackerrank.com...



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HackerRank Question - In HackerRank.com/test/g1lj816k4tp/questions/8aghtpsmcb

ORACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 8 / 36 01 hour 20 mins Rohan Dobarkar

ALL

19

20

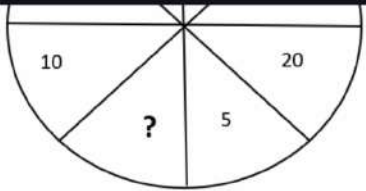
21

22

23

24

25



Pick ONE option

- ☒ has been suffering
- ☐ has suffered
- ☐ is suffer
- ☐ No improvement

Clear Selection

Mac dock icons: Safari, Messages, Mail, Photos, Calendar, App Store, Music, Podcasts, TV, Google Chrome, Word, Excel, Trash

ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HackerRank Question - V x +

hackerrank.com/test/g1j816k4tp/questions/cefsjmet244

ORACLE

FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 7 / 36

01 hour 21 mins

Rohan Dobarkar

ALL

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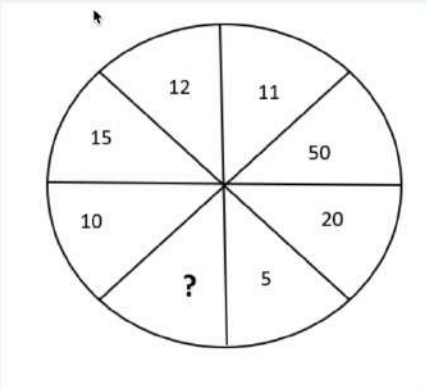
24

25

26

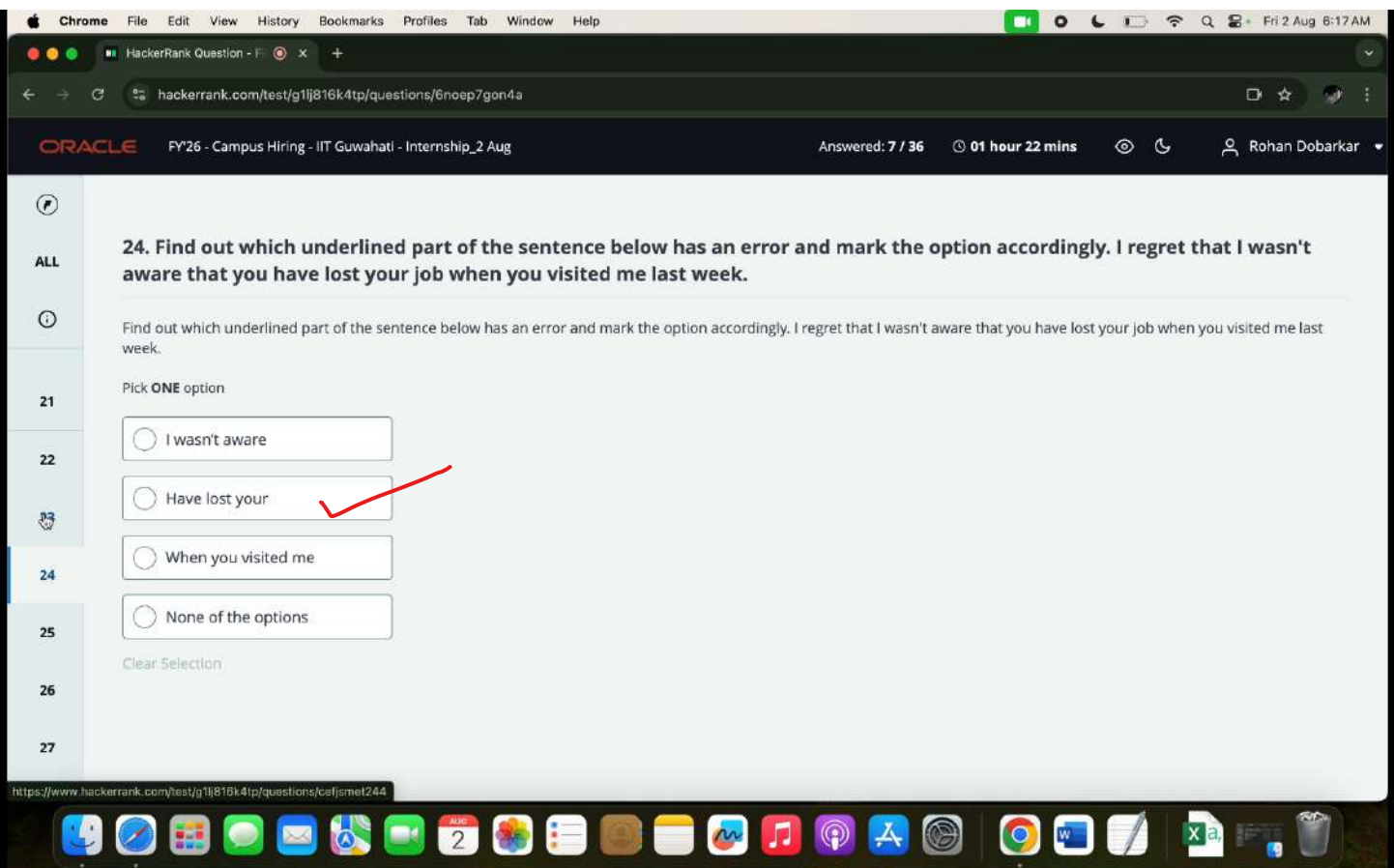
23. Which of the following four sentence is grammatically correct?

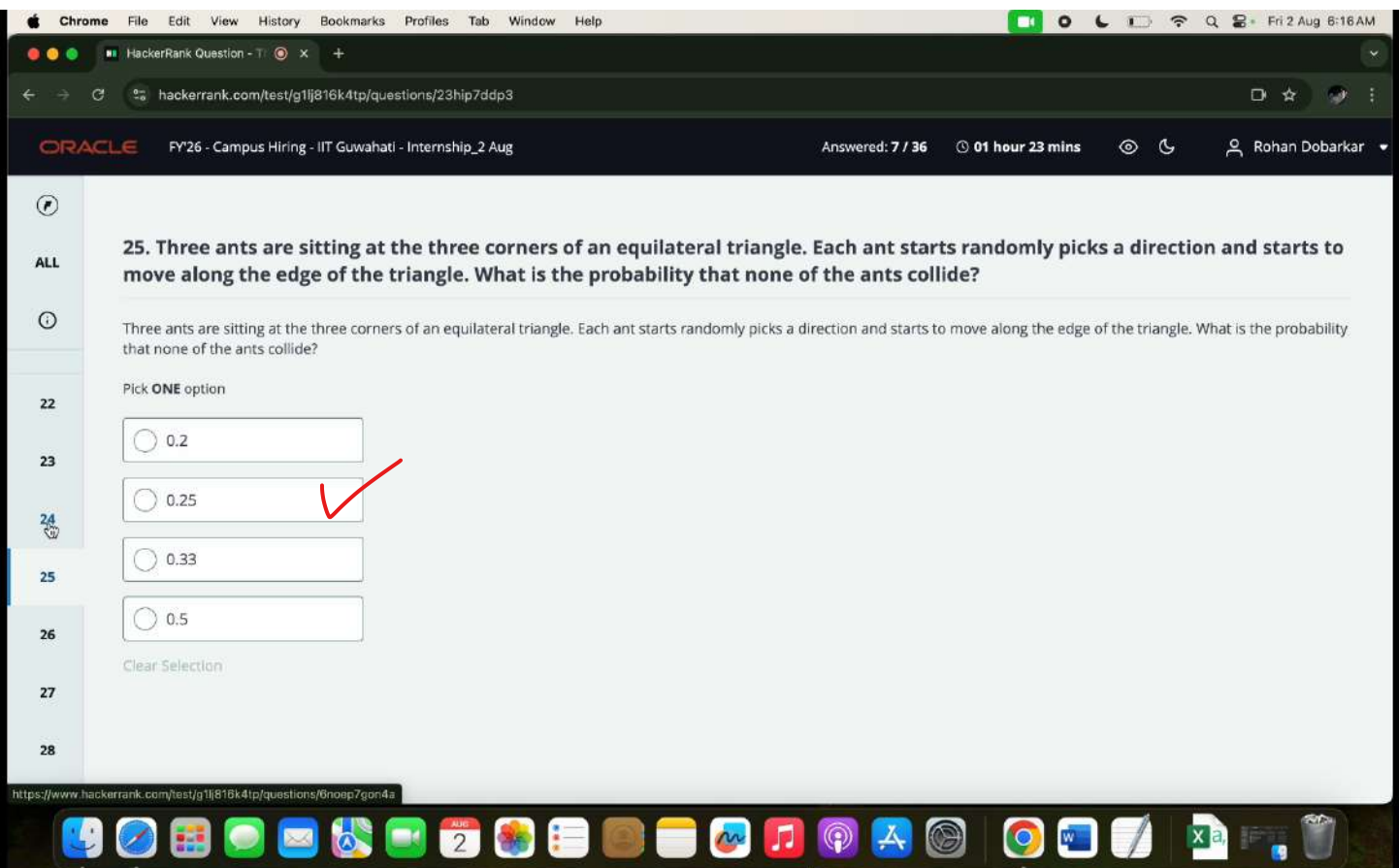
Which of the following four sentence is grammatically correct?



Pick **ONE** option







Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - A x +

hackerrank.com/test/g1lj816k4tp/questions/39jt3krog4d

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 6 / 36 01 hour 24 mins Rohan Dobarkar

ALL

26. A regular clock has an hour and minute hand. At 12 midnight the hands are exactly aligned. How many times a day will they overlap?

A regular clock has an hour and minute hand. At 12 midnight the hands are exactly aligned. How many times a day will they overlap?

Pick **ONE** option

23 ☐ 24

24 ☐ 23

25 ☒ 22

26 ☐ 21

27 Clear Selection

28

✓



Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - T x

hackerrank.com/test/g1lj816k4tp/questions/9sok2najkjd

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 6 / 36 01 hour 25 mins Rohan Dobarkar

27. There is a building of 100 floors -If an egg drops from the Nth floor or above it will break. -If it's dropped from any floor below, it will not break. You're given 2 eggs. How many drops you need to make?

There is a building of 100 floors
-If an egg drops from the Nth floor or above it will break.
-If it's dropped from any floor below, it will not break.
You're given 2 eggs.
How many drops you need to make?

Pick ONE option

12

14

17

20

Clear Selection

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - T x

hackerrank.com/test/g1lj816k4tp/questions/8h4po7bj7ed

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 5 / 36 01 hour 25 mins Rohan Dobarkar

28. There are 9 coins. Out of which one is odd one i.e weight is less. How many iterations of weighing are required to find odd coin?

There are 9 coins. Out of which one is odd one i.e weight is less. How many iterations of weighing are required to find odd coin?

Pick **ONE** option

25 ☐ 2

26 ☒ 3 ✓

27 ☐ 4

28 ☐ 5

Clear Selection

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - W x +

hackerrank.com/test/g1lj816k4tp/questions/fjnad84

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 4 / 36 01 hour 26 mins Rohan Dobarkar

29. What is the remainder when 20001000 divided by 13?

What is the remainder when 2000^{1000} divided by 13?

Pick **ONE** option

26 ☐ 3 ✓

27 ☐ 4

28 ☐ 8

29 ☐ 11

Clear Selection

30. Interpret Charts

09:52 AM 06-08-2024

30. Interpret Charts

ALL



27

28

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30

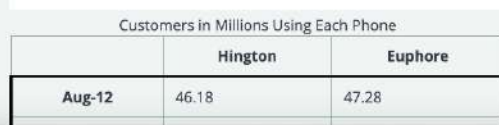
31

✓

33

12

The following graph shows the number of people (in millions) using Hington and Euphore brand phones in France from August to October 2012. Only these two brands of phones exist in the market.



ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HackerRank Question - In HackerRank.com/test/g1j816k4tp/questions/75eqgqdkj3o

OracleFY26 - Campus Hiring - IIT Guwahati - Internship_2 AugAnswered: 3 / 3601 hour 32 minsRohan Dobarkar

ALL

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33

Customers in Millions Using Each Phone

	Hington	Euphore
Aug-12	46.18	47.28
Sep-12	47.34	48.91
Oct-12	49.32	49.27

The following chart shows the addition of customers (in millions) of Hington and Euphore in each of the three months. There are two versions of the Euphore brand phones, Version-1, and Version-2.

Addition of customers (in millions)

Hington

Version-1

Version-2

	Hington	Version-1	Version-2
Aug-12	0.5	1.34	0.5
Sep-12	0.5	1.34	0.51
Oct-12	0.5	1.34	0.51

Addition of Customers in millions

	Hington	Version-1	Version-2
--	---------	-----------	-----------

The screenshot shows a HackerRank test interface. At the top, the browser address bar displays the URL: `hackerrank.com/test/g1j816k4tp/questions/75eqgdk3o`. The header bar includes the Oracle logo, the test title "FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug", and the user's progress: "Answered: 3 / 36" and "01 hour 31 mins". The user's name, "Rohan Dobarkar", is also visible.

The main content area displays a table titled "Addition of Customers in millions". The table has four columns: "Month", "Hington", "Version-1", and "Version-2". The data rows are for August-12, September-12, and October-12.

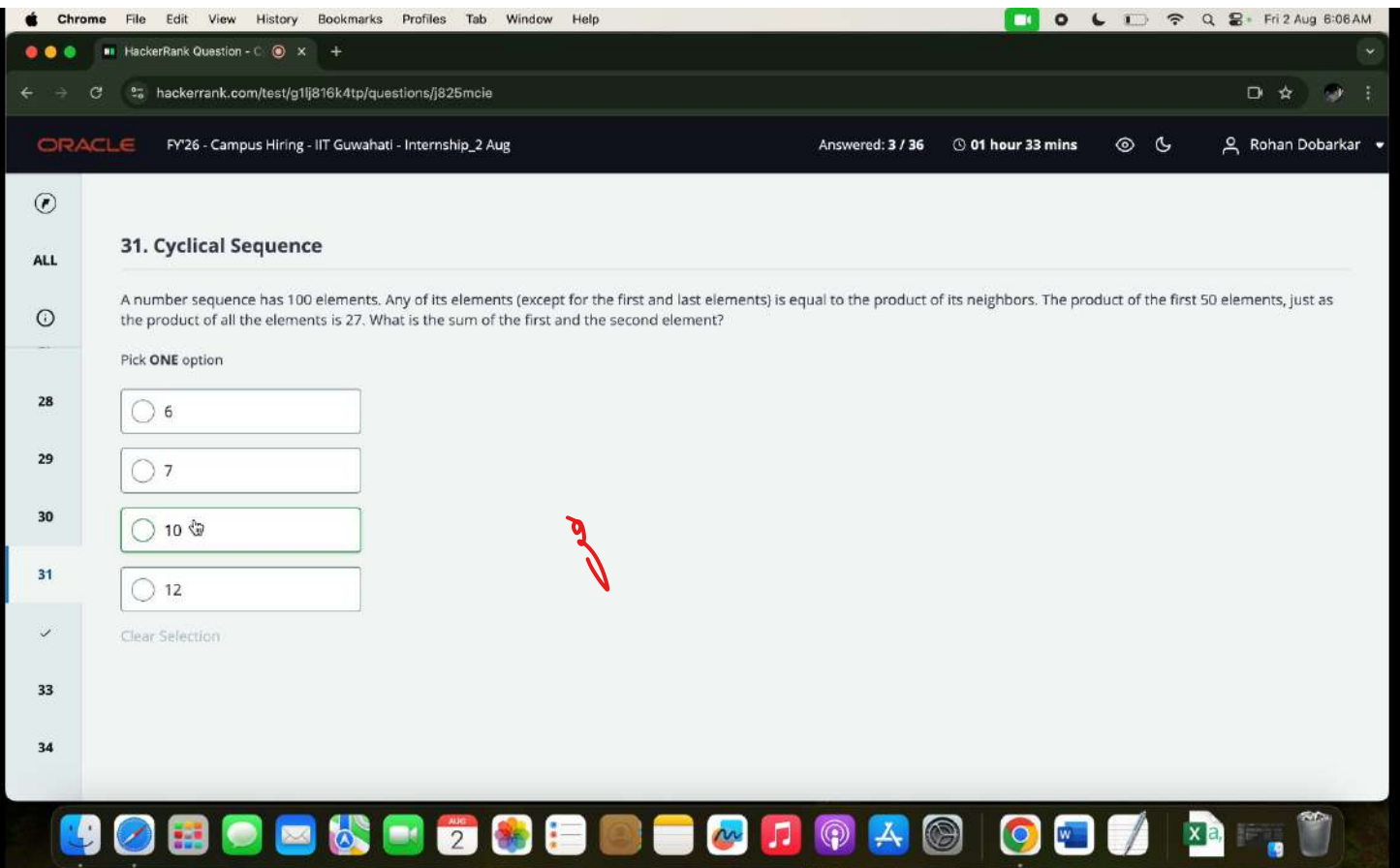
Month	Hington	Version-1	Version-2
Aug-12	0.36	1.24	0.6
Sep-12	1.16	1.08	0.8
Oct-12	1.98	0.84	0.48

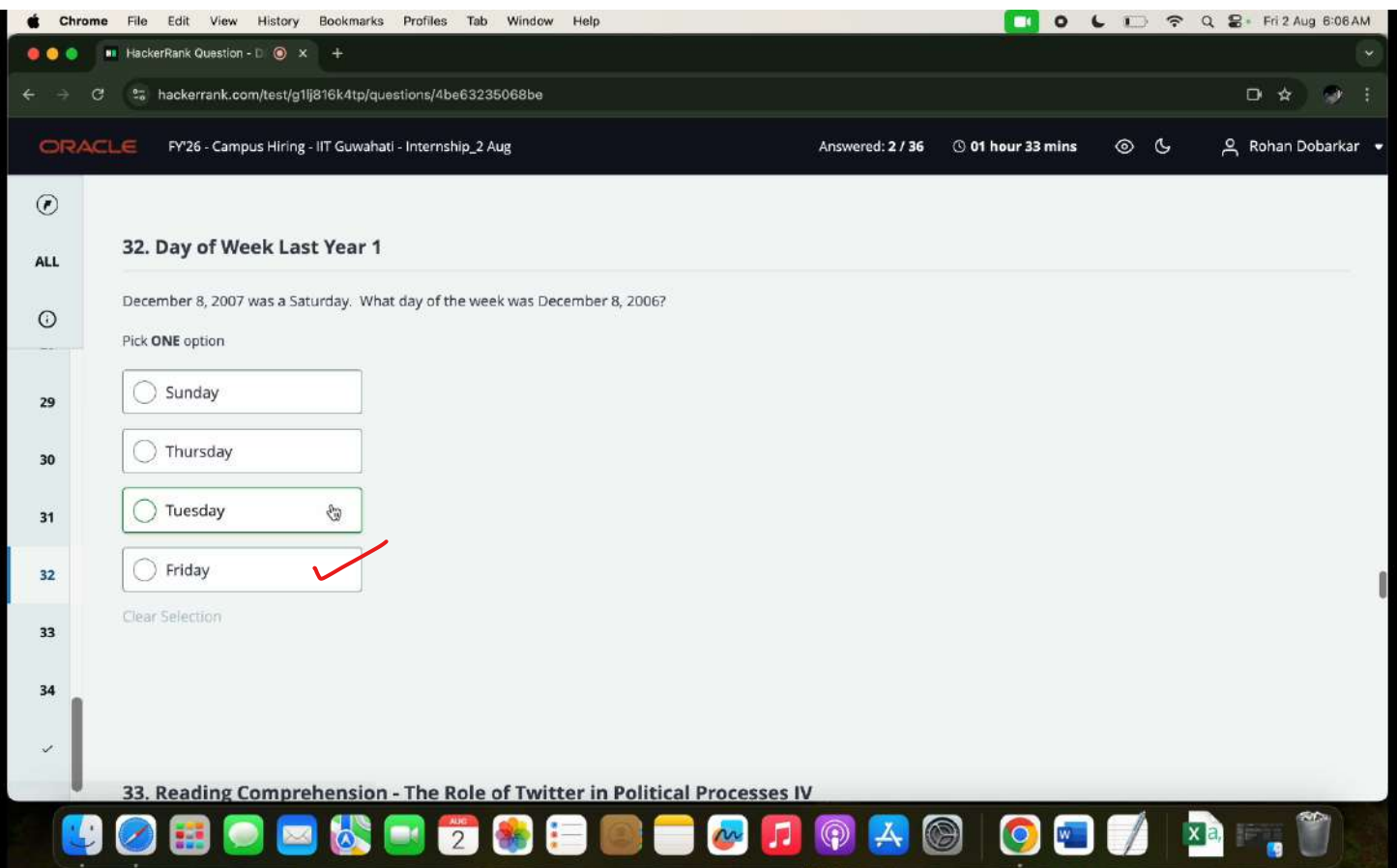
Below the table, a question is asked: "What was the total number of people using either of the two brands in July 2012?". The user is instructed to "Pick ONE option".

The options provided are:

- ☐ 91.21 million
- ☐ 6.87 million
- ☐ 98.23 million
- ☐ None of these

At the bottom of the question area, there is a "Clear Selection" link.





ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HackerRank Question - R

hackerrank.com/test/g1lj816k4tp/questions/4tdh7rn3593

Fri 2 Aug 6:06 AM

ORACLEFY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 2 / 3601 hour 33 minsRohan Dobarkar

ALL

29

30

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32

33

34

✓

✓

33. Reading Comprehension - The Role of Twitter in Political Processes IV

Read the passage and answer the questions that follow.

"Just putting up my twttr," the first tweet ever made by Twitter co-founder Jack Dorsey, was posted in 2006. It signaled the platform's usage as a tool for global social communication and to compete with sites like Facebook and My Space.

Since that landmark post went up more than sixteen years ago, Twitter has completely changed how information is gathered and disseminated. Everyone around the world who has a mobile phone or laptop can sign up for free. They can also easily keep abreast of happenings through tweets carried across the different pages of its users.

Whether knowingly or inadvertently, the discourse on Twitter now frequently incorporates key political elements in various forms. This is seen in its widespread use by politicians as a quick, simple, and inexpensive method of public outreach.

A single post may reach the same number of people just as rapidly as a TV commercial. It is also widely known to cost a minuscule fraction of the cost of TV advertising. Twitter's mandated brevity further ensures more users read the entire message because a 140-character remark only takes a few seconds to read and retweet.

In contrast to conventional methods of communication, feedback is produced. Any high-profile government official can fully access the unfiltered emotions of a segment of the populace. It has also come to function as the online version of town halls. People are no longer afraid to air their grievances directly to those who are in authority and can bring about the desired change.

Therefore, it should come as no surprise that Twitter has played a significant role in the development of many movements. These movements provide a united voice for people with regards to socio-political issues. As such, new legislation or changes to old laws have been enacted more quickly.

The 2017 Me-Too campaign against sexual abuse, sexual harassment, and rape culture is a fitting case study. Its impact reverberated across the world and is still discussed in many hallowed chambers.

One of its far-reaching effects on political processes was the Member and Employee Training and Oversight on Congress Act (ME TOO Congress Act) on November 15, 2017, proposed by US legislator Jackie Speier. Under the old system, complaints regarding the legislative branch took months of counseling and mediation before it could be filed. That has now changed, and victims can now properly and quickly seek legal redress without having to worry about their jobs or backlash.

Beyond this, Twitter has also affected political processes in other unique ways, such as electioneering, be it passive or active. Campaigns take place on the platform like never before, contributing largely to the emergence of political social media influencers.

However, a lot of people have expressed concerns about the use of Twitter's data to gather and interpret citizen voices. Such concerns stem from the possibility that the preferences of citizens expressed through the platform may be more focused on polarizing political issues than on solving challenging problems.

Depending on who you ask, Twitter's enormous rise in accessibility to political processes may or may not be a positive thing. Whatever the case, there is no denying that it has taken on

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - R... x +

hackerrank.com/test/g1j816k4tp/questions/4tdh7rn3593

ORACLE FY'26 - Campus Hiring - IIT Guwahati - Internship_2 Aug

Answered: 2 / 36 01 hour 33 mins Rohan Dobarkar

ALL

29

30

31

32

33

34

✓

✓

The 2017 #MeToo campaign against sexual abuse, sexual harassment, and rape culture is a fitting case study. Its impact reverberated across the world and is still discussed in many hallowed chambers.

One of its far-reaching effects on political processes was the Member and Employee Training and Oversight on Congress Act (ME TOO Congress Act) on November 15, 2017, proposed by US legislator Jackie Speier. Under the old system, complaints regarding the legislative branch took months of counseling and mediation before it could be filed. That has now changed, and victims can now properly and quickly seek legal redress without having to worry about their jobs or backlash.

Beyond this, Twitter has also affected political processes in other unique ways, such as electioneering, be it passive or active. Campaigns take place on the platform like never before, contributing largely to the emergence of political social media influencers.

However, a lot of people have expressed concerns about the use of Twitter's data to gather and interpret citizen voices. Such concerns stem from the possibility that the preferences of citizens expressed through the platform may be more focused on polarizing political issues than on solving challenging problems.

Depending on who you ask, Twitter's enormous rise in accessibility to political processes may or may not be a positive thing. Whatever the case, there is no denying that it has taken on a more significant role in the operations of such a vital aspect of today's society.

According to the passage, Twitter only allows for only 140 characters per tweet. What edge do you think it gives the app over other social media applications that allow for more use of characters?

Pick **ONE** option

☐ Correction can be easily made to tweets before uploading them.

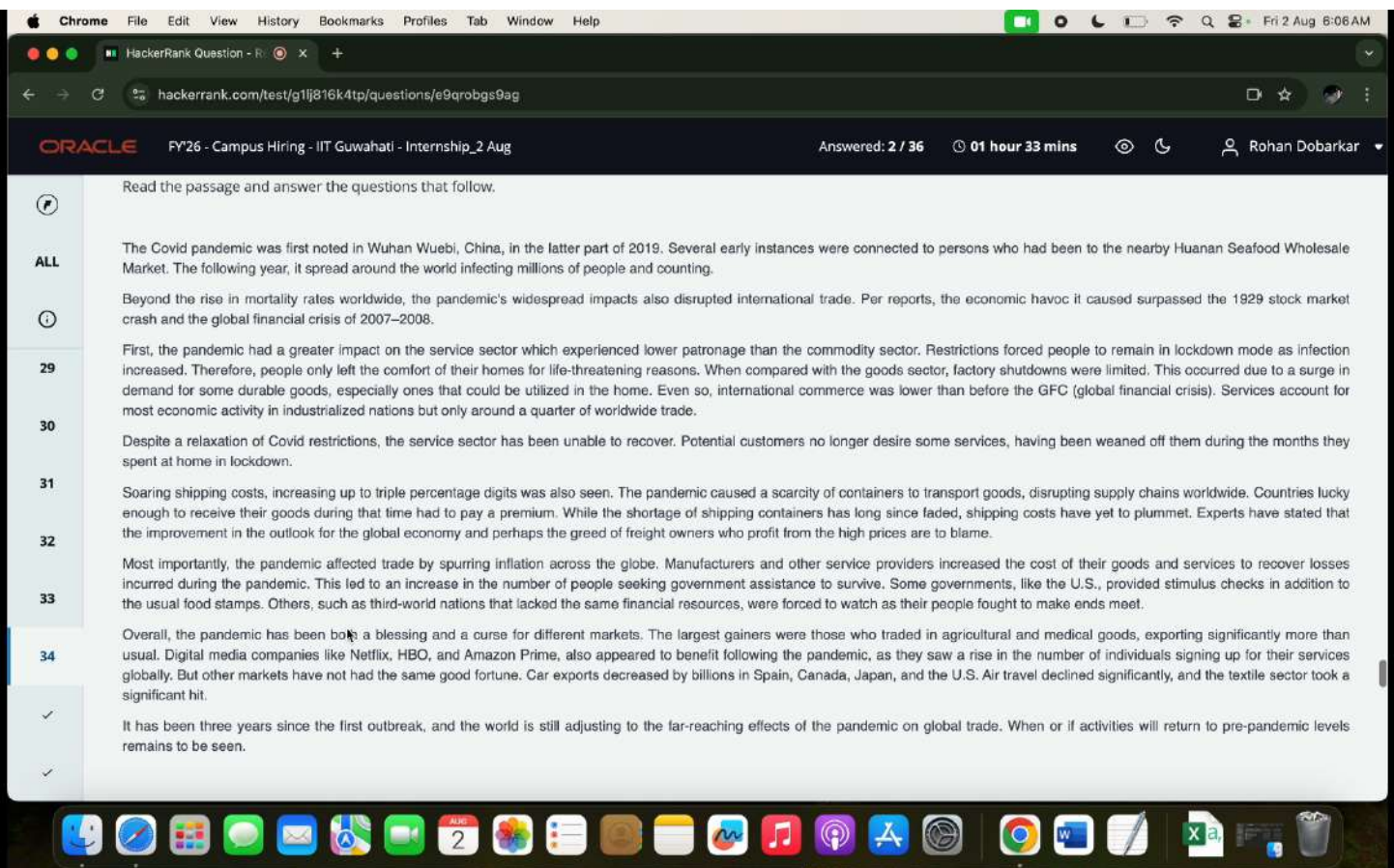
☐ It ensures that users can share their thoughts in little time even when busy.

☐ More people will get to consume information far more quickly.

☐ No other app can boast of having such a unique design.

Clear Selection

2



Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - R x +

hackerrank.com/test/g1lj816k4tp/questions/e9qrobg9ag

ORACLE FY26 - Campus Hiring - IIT Guwahati - Internship_2 Aug Answered: 2 / 36 01 hour 33 mins Rohan Dobarkar

Which of the following does not correctly utilize the phrase other similar terms to "weaned off" as used in the passage?

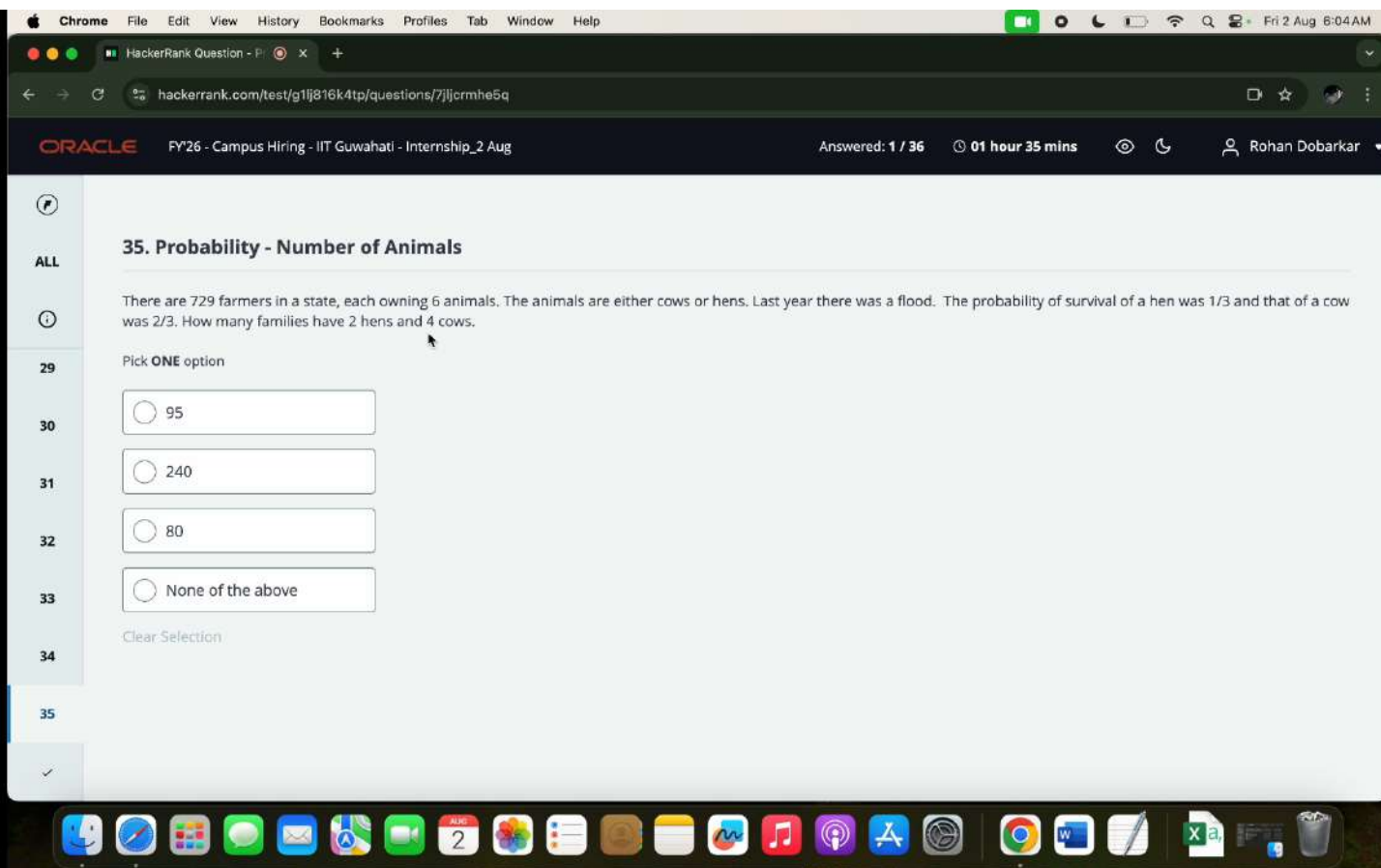
Pick ONE option

- ☐ Some seal pups are taken off their mothers' teats after only two weeks.
- ☐ I'm trying to encourage people to eat fruits and vegetables.
- ☒ The medication helps dissuade patients from taking alcohol by simulating its effects on the brain.
- ☐ I want to curtail my kids' interest in T.V. and video games

Clear Selection

35. Probability - Number of Animals

There are 729 farmers in a state, each owning 6 animals. The animals are either cows or hens. Last year there was a flood. The probability of survival of a hen was $\frac{1}{3}$ and that of a cow was $\frac{2}{3}$. How many families have 2 hens and 4 cows.



ChromeFileEditViewHistoryBookmarksProfilesTabWindowHelp

HackerRank Question - P...hackerrank.com/test/g1j816k4tp/questions/aaf1mqfg4hk

ORACLEFY26 - Campus Hiring - IIT Guwahati - Internship_2 AugAnswered: 0 / 3601 hour 37 minsRohan Dobarkar

36. Pollution Chart

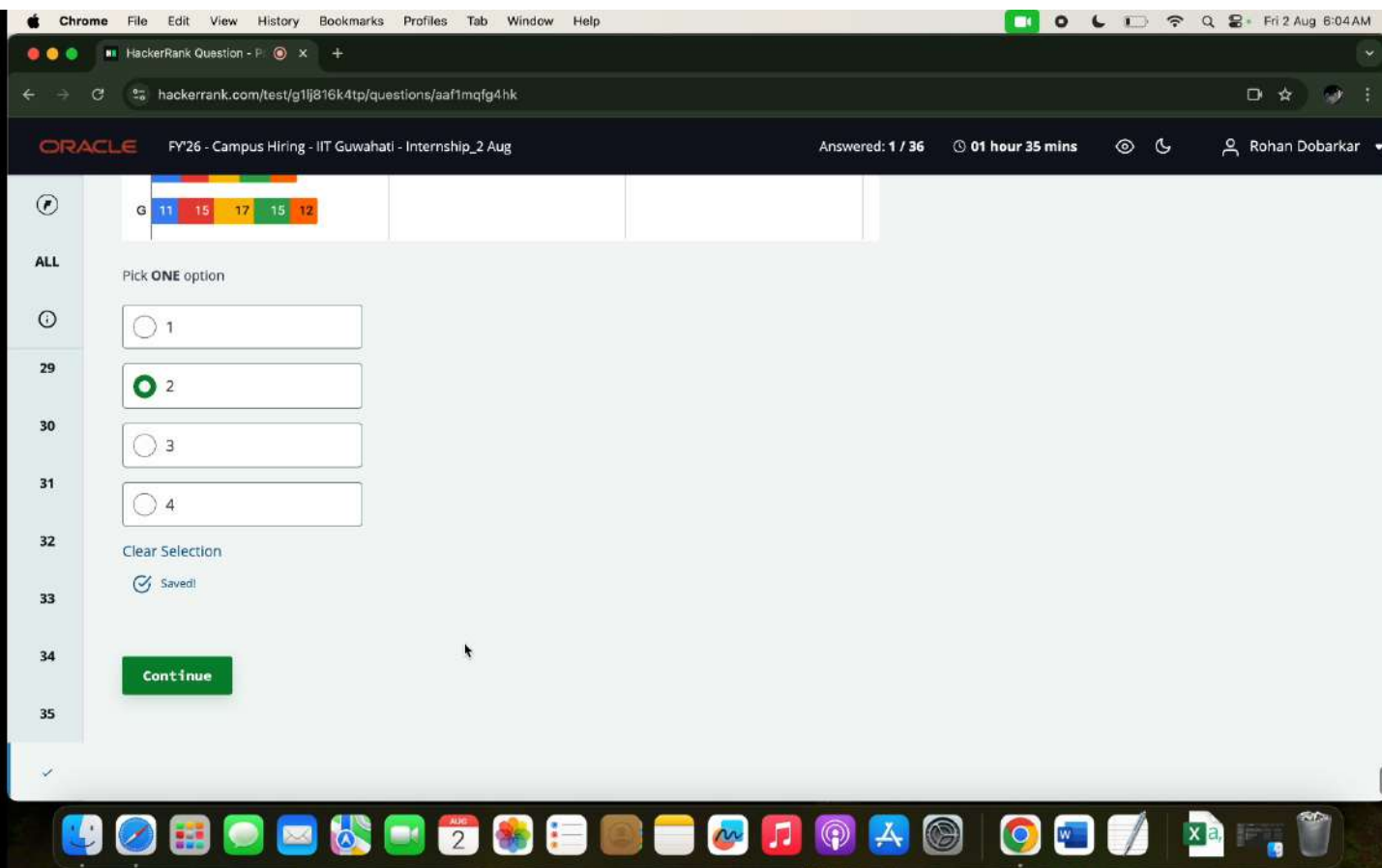
The graph shows the pollution index of 7 cities- A, B, C, D, E, F, and G from 2006 to 2010. If for each year, the cities are ranked in terms of ascending order of pollution index (the city with least pollution index stands first), then how many cities do not change their ranking more than once from 2006 to 2008.

2006, 2007, 2008, 2009 and 2010

City	2006	2007	2008	2009	2010
A	13	21	24	25	19
B	15	17	11	32	22
C	34	35	29	35	35
D	56	57	56	67	65
E	57	45	45	56	57
F	12	12	13	13	11
G	11	15	17	15	12

Pick ONE option

☐ 1



Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - x

hackerrank.com/test/g1j816k4tp/questions/apm6j4dps8d

1h 12m left

talent = [1, 2, 3, 2, 1]

There is no way fewer than 3 students can have at least 3 talents. The following subarrays of length 3 or more are possible:
Starting at position 1: [1, 2, 3], [1, 2, 3, 2], [1, 2, 3, 2, 1]
Starting at position 2: [2, 3, 2], [2, 3, 2, 1]
Starting at position 3: [3, 2, 1]
No further subarrays of length 3 or more exist.

1 Starting at the first position, the shortest subarray with one of each talent value has length 3.
✓ Starting at element 2, the shortest has length 4.
At position 3, the shortest and only possible subarray has length 3.
3 It is not possible to form a team with fewer than 3 members, so the subarrays starting at the last two positions will fail.
4
5 The highlighted subarrays are selected. The return array is [3, 4, 3, -1, -1].

6 **Function Description**
7 Complete the function `teamSize` in the editor below. It must return an array of integers.
8 `teamSize` has the following parameter(s):
talent: An array of length n where `talent[i]`

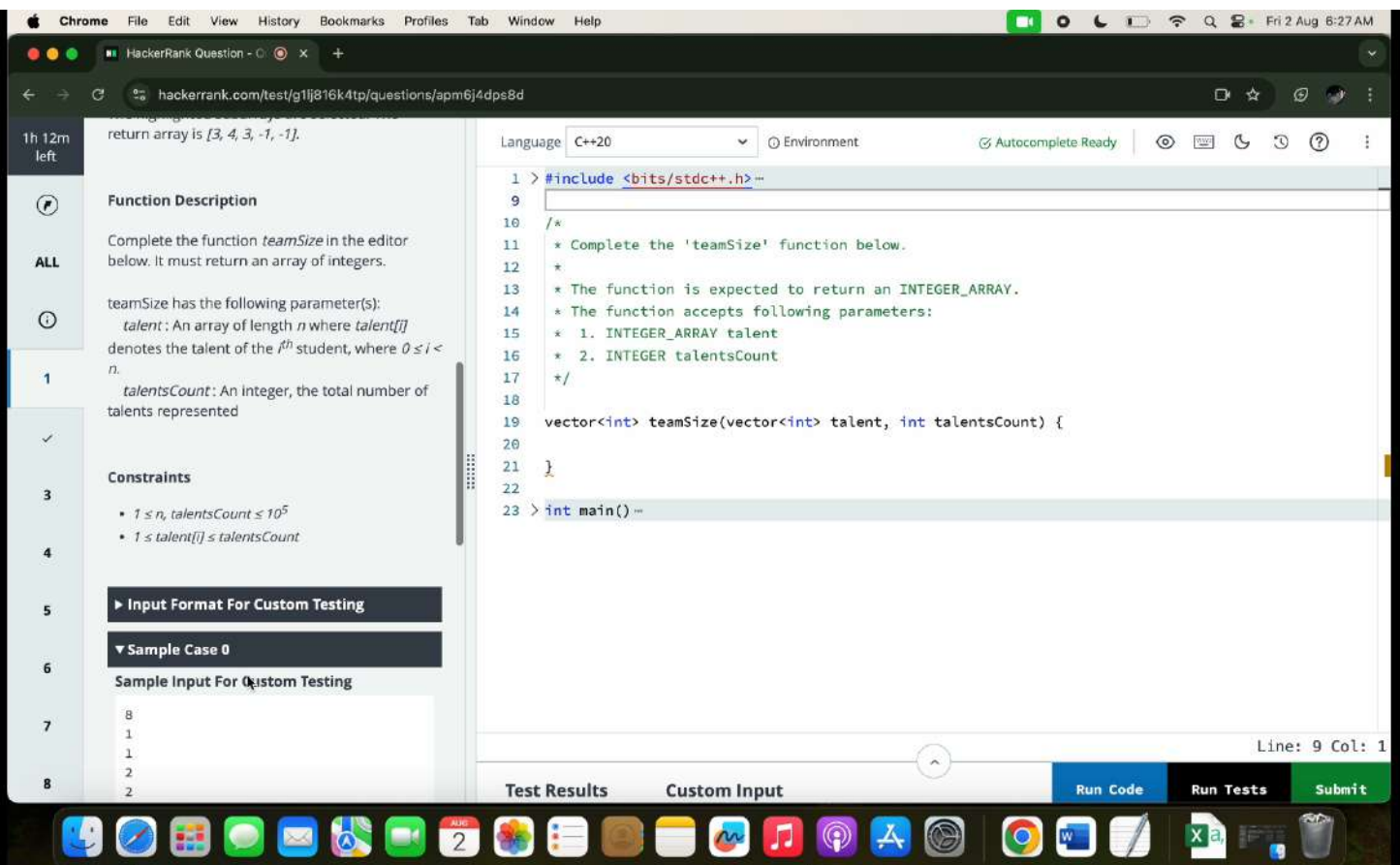
Language C++20 Environment Autocomplete Ready

```
1 > #include <bits/stdc++.h> --
9
10 /*
11  * Complete the 'teamSize' function below.
12  *
13  * The function is expected to return an INTEGER_ARRAY.
14  * The function accepts following parameters:
15  * 1. INTEGER_ARRAY talent
16  * 2. INTEGER talentsCount
17  */
18
19 vector<int> teamSize(vector<int> talent, int talentsCount) {
20
21 }
22
23 > int main() --
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

2



Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - x +

hackerrank.com/test/g1j816k4tp/questions/apm6j4dps8d

1h 12m left

• $1 \leq \text{talent}[i] \leq \text{talentsCount}$

▶ Input Format For Custom Testing

▼ Sample Case 0

Sample Input For Custom Testing

```
8
1
1
2
2
3
1
3
2
2
3
```

Sample Output

```
5
4
4
3
4
3
-1
-1
```

Explanation

$n = 8$
 $\text{talent} = [1, 1, 2, 2, 3, 1, 3, 2]$
 $\text{talentsCount} = 3$

Language C++20 Environment Autocomplete Ready

```
1 > #include <bits/stdc++.h> ...
9
10 /*
11  * Complete the 'teamSize' function below.
12  *
13  * The function is expected to return an INTEGER_ARRAY.
14  * The function accepts following parameters:
15  * 1. INTEGER_ARRAY talent
16  * 2. INTEGER talentsCount
17  */
18
19 vector<int> teamSize(vector<int> talent, int talentsCount) {
20
21 }
22
23 > int main() ...
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

2

Chrome File Edit View History Bookmarks Profiles Tab Window Help

HackerRank Question - HackerRank.com/test/g1lj816k4tp/questions/apm6j4dps8d

1h 12m left

3
-1
-1

Explanation
 $n = 8$
 $talent = [1, 1, 2, 2, 3, 1, 3, 2]$
 $talentsCount = 3$

ALL

1 There have to be at least 3 elements in any subarray to represent each talent. The following subarrays are possible:
Beginning at the first position: [1, 1, 2], [1, 1, 2, 2], [1, 1, 2, 3], [1, 1, 2, 3, 1], [1, 1, 2, 3, 1, 3], [1, 1, 2, 3, 1, 3, 2]

2 Second position: [1, 2, 2], [1, 2, 2, 3], [1, 2, 2, 3, 1], [1, 2, 2, 3, 1, 3], [1, 2, 2, 3, 1, 3, 2]

3 Third position: [2, 2, 3], [2, 2, 3, 1], [2, 2, 3, 1, 3], [2, 2, 3, 1, 3, 2]

4 Fourth position: [2, 3, 1], [2, 3, 1, 3], [2, 3, 1, 3, 2]

5 Fifth position: [3, 1, 3], [3, 1, 3, 2]

6 Sixth position: [1, 3, 2]

7 No further subarrays will have 3 elements.

8 The highlighted subarrays are the shortest for each starting position. The return array of subarray lengths is [5, 4, 4, 3, 4, 3, -1, -1]

► Sample Case 1

► Sample Case 2

Language C++20 Environment Autocomplete Ready

```
1 > #include <bits/stdc++.h> --
9
10 /*
11  * Complete the 'teamSize' function below.
12  *
13  * The function is expected to return an INTEGER_ARRAY.
14  * The function accepts following parameters:
15  * 1. INTEGER_ARRAY talent
16  * 2. INTEGER talentsCount
17  */
18
19 vector<int> teamSize(vector<int> talent, int talentsCount) {
20
21 }
22
23 > int main() --
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

2

1h 12m left

ALL

1

✓

3

4

5

6

7

8

The highlighted subarrays are the shortest for each starting position. The return array of subarray lengths is [5, 4, 4, 3, 4, 3, -1]

▼ Sample Case 1

Sample Input For Custom Testing

```
5
1
1
1
1
1
1
```

Sample Output

```
1
1
1
1
1
```

Explanation

$n = 5$
 $talent = [1, 1, 1, 1, 1]$
 $talentsCount = 1$

Since there is only 1 type of talent, any 1 person can form a team.

► Sample Case 2

Language C++20 Environment Autocomplete Ready

```
1 > #include <bits/stdc++.h> ~
9
10 /*
11  * Complete the 'teamSize' function below.
12  *
13  * The function is expected to return an INTEGER_ARRAY.
14  * The function accepts following parameters:
15  * 1. INTEGER_ARRAY talent
16  * 2. INTEGER talentsCount
17  */
18
19 vector<int> teamSize(vector<int> talent, int talentsCount) {
20
21 }
22
23 > int main() ~
```

Line: 9 Col: 1

Test Results Custom Input Run Code Run Tests Submit

1h 12m left

ALL

1

4
7

Sample Output

8
7
-1
-1
-1
-1
-1
-1

Explanation
 $n = 9$
 $talent = [7, 5, 3, 4, 6, 1, 7, 2, 4]$
 $talentsCount = 7$

3
4
5
6
7
8

There have to be at least 7 elements in any subarray to represent each talent. The following subarrays are possible:

Beginning at the first position: $[7, 5, 3, 4, 6, 1, 7]$, $[7, 5, 3, 4, 6, 1, 7, 2]$, $[7, 5, 3, 4, 6, 1, 7, 2, 4]$

Second position: $[5, 3, 4, 6, 1, 7, 2]$, $[5, 3, 4, 6, 1, 7, 2, 4]$

Third position: $[3, 4, 6, 1, 7, 2, 4]$

No further subarrays will have 7 elements.

The highlighted subarrays are the shortest for each starting position. The return array of subarray lengths is $[8, 7, -1, -1, -1, -1, -1, -1]$.

Language C++20 Environment Autocomplete Ready

```
1 > #include <bits/stdc++.h> ...
9
10 /*
11  * Complete the 'teamSize' function below.
12  *
13  * The function is expected to return an INTEGER_ARRAY.
14  * The function accepts following parameters:
15  * 1. INTEGER_ARRAY talent
16  * 2. INTEGER talentsCount
17  */
18
19 vector<int> teamSize(vector<int> talent, int talentsCount) {
20
21 }
22
23 > int main() ...
```

Test Results Custom Input Run Code Run Tests Submit

Line: 9 Col: 1

ALL

Determine if a triangle formed by three points $a(x_1, y_1)$, $b(x_2, y_2)$, and $c(x_3, y_3)$ is non-degenerate. If two given points, $p = (x_p, y_p)$ and $q = (x_q, y_q)$, are located on or inside a triangle, they belong to the triangle. Return the correct scenario number.

Scenarios:

- 1
 - 0: the lines do not form a valid non-degenerate triangle
 - 1: point p belongs to the triangle but point q does not
- 2
 - 2: point q belongs to the triangle but point p does not
 - 3: both points p and q belong to the triangle
- 3
 - 4: neither point p nor point q belong to the triangle

Note: A triangle is considered non-degenerate if it meets the following conditions, where $|ab|$ denotes the length of the line segment between points a and b :

- $|ab| + |bc| > |ac|$
- $|bc| + |ac| > |ab|$
- $|ab| + |ac| > |bc|$

Example

```
1 =  
a(x1, y1):  
(2, 2)  
2 =  
b(x2, y2):
```

```
double area(int x1, int y1, int x2, int y2, int x3, int y3) {  
    return fabs(x1*(y2-y3) + x2*(y3-y1) + x3*(y1-y2)) / 2.0;  
}
```



ALL

Note: A triangle is considered non-degenerate if it meets the following conditions, where $|ab|$ denotes the length of the line segment between points a and b:

①

- $|ab| + |bc| > |ac|$
- $|bc| + |ac| > |ab|$
- $|ab| + |ac| > |bc|$

1

2

Example

3

1 =

a(x1,y1):

(2,2)

4

2 =

b(x2,y2):

(7,2)

5

3 =

c(x3,y3):

(5,4)

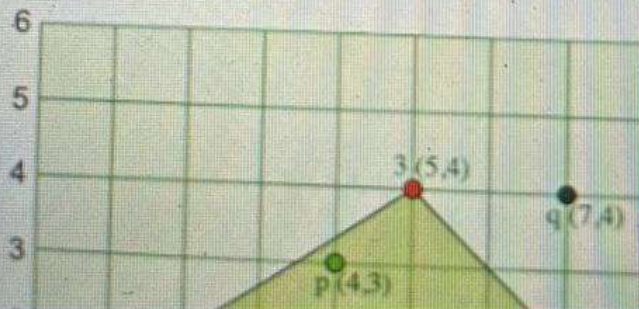
6

p = p(xp,

yp): (4,3)

q = q(xq,

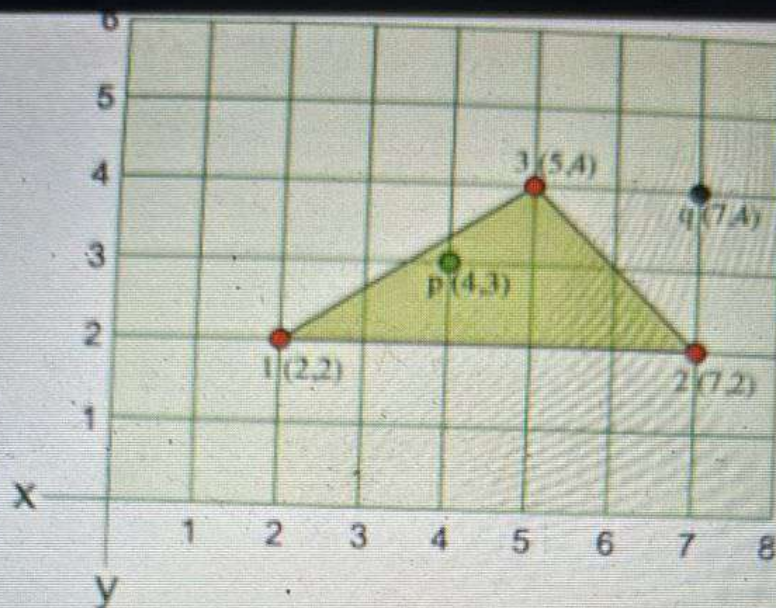
yq): (7,4)



```
bool IsInside(int x1, int y1, int x2, int y2, int x3,  
int y) {
```

Search





First, the triangle abc forms a valid non-degenerate triangle.

- $|ab| = 7 - 2 = 5$. $|bc| = \sqrt{(7-5)^2 + (4-2)^2} = \sqrt{2^2 + 2^2} = \sqrt{8} = 2.82$. $|ac| = \sqrt{(5-2)^2 + (4-2)^2} = \sqrt{3^2 + 2^2} = \sqrt{13} = 3.6$.
- $|ab| + |bc| > |ac| \Rightarrow 5 + 2.82 > 3.6$
- $|bc| + |ac| > |ab| \Rightarrow 2.82 + 3.6 > 5$
- $|ab| + |ac| > |bc| \Rightarrow 5 + 3.6 > 2.82$

Second, the point p(5, 4) belongs to the triangle but point q(7, 4) does not as shown. So, the answer is 1.

Function Description

Complete the function `pointsBelong` in the editor below.

`pointsBelong` has the following parameter(s):

- $|ab| = 7 - 2 = 5$. $|bc| = \sqrt{(7-5)^2 + (4-2)^2} = \sqrt{2^2 + 2^2} = \sqrt{8} = 2.82$. $|ac| = \sqrt{(5-2)^2 + (4-2)^2} = \sqrt{3^2 + 2^2} = \sqrt{13} = 3.6$.
- $|ab| + |bc| > |ac| \Rightarrow 5 + 2.82 > 3.6$
- $|bc| + |ac| > |ab| \Rightarrow 2.82 + 3.6 > 5$
- $|ab| + |ac| > |bc| \Rightarrow 5 + 3.6 > 2.82$

Second, the point $p(5, 4)$ belongs to the triangle but point $q(7, 4)$ does not as shown. So, the answer is 1.

Function Description

Complete the function `pointsBelong` in the editor below.

`pointsBelong` has the following parameter(s):

`int x1, y1, x2, y2, x3, y3`: integer coordinates of the three points that may create a valid triangle

`int xp, yp, xq, yq`: integer coordinates of the two points p and q

Returns

`int`: the scenario number

Constraints

- $0 \leq x1, y1, x2, y2, x3, y3, xp, yp, xq, yq \leq 2000$

► Input Format for Custom Testing

▼ Sample Case 0



returns

int: the scenario number

Constraints

- $0 \leq x1, y1, x2, y2, x3, y3, xp, yp, xq, yq \leq 2000$

► Input Format for Custom Testing

▼ Sample Case 0

Sample Input 0

STDIN	Function
0	→ (x1,y1) = (0,0)
0	
2	→ (x2,y2) = (2,0)
0	
4	→ (x3,y3) = (4,0)
0	
2	→ p = (xp,yp) = (2,0)
0	
4	→ q = (xq,yq)=(4,0)
0	

Sample Output 0

0

Explanation 0

5



2. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

☒ This memory address does not map to actual memory cell that corresponds to address number.

☐ This memory is the same as physical memory.

Solve question 3

☐ This memory cannot be accessed by applications at run time.

☐ This memory address is always present even after a system is restarted.

Clear Selection

3. Consider the two functions incr and decr shown below. incr(){ decr(){ wait(s
are 5 threads each invoking incr once, and 3 threads each invoking decr once
shared variable X is 10 Suppose there are two implementations of the sema
initialized to 1, 1, 2. S is a counting semaphore initialized to 2. Let V1 and V2 b

4. What is the main difference between "user mode" and "kernel mode"?

What is the main difference between "user mode" and "kernel mode" in an operating system?

Pick **ONE** option

- ☐ User mode allows direct hardware access, kernel mode does not.
- ☒ Kernel mode can access protected memory regions, user mode cannot.
- ☐ Kernel mode applications are given less priority than user mode applications.
- ☐ User mode instructions can switch the CPU to kernel mode, but not vice-versa.

Clear Selection

5. Which data structure is best suited for implementing a priority queue?

Which data structure is best suited for implementing a priority queue?

<https://www.hackerrank.com/test/g1lj816k4tp/questions/c3aaf70q0e6>

ALL



1

2

3

4



6

7

8

9

5. Which data structure is best suited for imp

Which data structure is best suited for implementing a priority c

Pick **ONE** option

☐ Array

☐ Slack

☒ Heap ✓

☐ Linked List

Clear Selection

6. In the context of Dynamic Programming,

In the context of Dynamic Programming, what is "memoizatio

<https://www.hackerrank.com/test/g1lj816k4tp/questions/a7dn8s3ffm6>

In the context of Dynamic Programming, what is "memoization"?

Pick **ONE** option

- ☐ Writing an algorithm in a memo to remember it
- ☐ Recomputing solutions to subproblems multiple times for accuracy
- ☐ Storing solutions to subproblems to avoid recomputation
- ☒ Optimizing a problem by breaking it into subproblems and solving iteratively

Clear Selection

7. A certain program must store the names of all the people who finish the finish line as the competitors finish. After the race is over the program must output the names of the competitors who placed 1 to 10 and then 100, 200, 300 and so on, as the competitors finish. What data structure would be the best choice for this program?

<https://www.hackerrank.com/test/g1lj816k4tp/questions/asnrkim1c2c>

```
#include <stdio.h>
int main() { char arr[] = "abcd"; char *p = arr;
printf("%c\n", *p); return 0; }
```

```
#include <stdio.h>
int main() {
    char arr[] = "abcd";
    char *p = arr;
    printf("%c\t", **p);
    printf("%c\t", *p++);
    printf("%c\t", (*p)++);
    printf("%c\n", *p);
    return 0;
}
```

Pick **ONE** option

☒ b b b c ✓

☐ b c c d

☐ b b c c

☐ b c c c

Clear Selection

<https://www.hackerrank.com/test/g1lj816k4tp/questions/3g83a54sl0o>

10. What is the latest long-term release of Oracle Database?

What is the latest long-term release of Oracle Database?

Pick **ONE** option

☒ 23ai

☐ 22ai

☐ 24ai

☐ 19ai

Clear Selection

11. Which of the following is not a property of

Which of the following is not a property of Relational Database

<https://www.hackerrank.com/test/g1lj816k4tp/questions/1qed2r1nbfi>

2. Which SQL constraint do we use to set some value to a table when value is not null?

Which SQL constraint do we use to set some value to a table when value is not null?

Pick **ONE** option

☐ NOT NULL

☒ DEFAULT

☐ CHECK

☐ UNIQUE

Clear Selection

13. Truncate table is classified as:

Truncate table is classified as:

www.hackerrank.com/test/g1lj816k4tp/questions/1pdsi1nm44h

1. How will you distribute incoming traffic to a set of Web Servers?

How will you distribute incoming traffic to a set of Web Servers?

Pick **ONE** option

- ☐ By deploying a Firewall before the servers
- ☐ By deploying a router before the servers.
- ☐ By deploying a load balancer before the servers.
- ☐ By deploying a switch before the servers.

Clear Selection

15. In a distributed system, what is CAP theorem?

In a distributed system, what is CAP theorem?

15. In a distributed system, what is CAP theorem?

In a distributed system, what is CAP theorem?

Pick **ONE** option

- ☐ It specifies the maximum allowable response time for a system
- ☐ It describes the trade-offs between consistency, availability, and partition tolerance
- ☐ It defines the minimum number of servers required for fault tolerance
- ☐ It outlines the process for scaling a system horizontally

Clear Selection

16. What is a Zombie process in Linux?

What is a Zombie process in Linux?

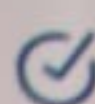
What is a Zombie process in Linux?

What is a Zombie process in Linux?

Pick ONE option

- ☐ A process which is running in the background.
- ☐ A process which is running for a long time.
- ☒ A process whose execution has completed but it still has an entry in the process table.
- ☐ A process which has crashed due to memory corruption.

Clear Selection

 Saved!

17. Which HTTP method is suitable for resource creation?

Which HTTP method is suitable for resource creation?

Pick ONE option

Which HTTP method is suitable for resource creation?

Which HTTP method is suitable for resource creation?

Choose **ONE** option

☐ PUT

☐ PATCH

☒ POST

☐ CREATE

Clear Selection

18. Which response status code category represents

29. What is the remainder when 2000^{1000} divided by 13?

What is the remainder when 2000^{1000} divided by 13?

Pick **ONE** option

☐ 3

☐ 4

☐ 8

☐ 11

Clear Selection

30. Interpret Charts

The following graph shows the number of people (in millions) using Hingto the market

<https://www.hackerrank.com/test/g1lj816k4tp/questions/75eqgqdk30>

1. Rearrange Students

In your school, two lines of students are formed A and B, with each line having exactly N students. The two lines of students will face each other. The Physical Education teacher wants the heights of all students standing across from each other to be equal. To do this, students may be swapped from one line to another. Any student in one line can be swapped with any student in the other line. Each swap counts as one operation, and each operation costs the minimum height in the swap. Reordering the students within one line has no cost. Determine the minimal cost to achieve the desired result. If it is impossible, return -1.

Example

$arrA = [4, 2, 2, 2]$

$arrB = [1, 4, 1, 2]$

If the arrays are sorted descending, they are $arrA = [4, 2, 2, 2]$ and $arrB = [4, 2, 1, 1]$. Swap a 2 in $arrA$ with a 1 in $arrB$ for a cost of $\min(1, 2) = 1$. This is the only swap that needs to occur so the answer is 1.

Function Description

Complete the function `rearrangeStudents` in the editor below.

`rearrangeStudents` has the following parameters:

- `int arrA[n]`: the heights of children in line A
- `int arrB[n]`: the heights of children in line B

ALL



1

2

3

4

5

6

7

8

9

In your school, two lines of students are formed A and B, with each line having exactly N students. The two lines of students will face each other. The Physical Education teacher wants the heights of all students standing across from each other to be equal. To do this, students may be swapped from one line to another. Any student in one line can be swapped with any student in the other line. Each swap counts as one operation, and each operation costs the minimum height in the swap. Reordering the students within one line has no cost. Determine the minimal cost to achieve the desired result. If it is impossible, return -1.

Example

$arrA = [4, 2, 2, 2]$

$arrB = [1, 4, 1, 2]$

If the arrays are sorted descending, they are $arrA = [4, 2, 2, 2]$ and $arrB = [4, 2, 1, 1]$. Swap a 2 in $arrA$ with a 1 in $arrB$ for a cost of $\min(1, 2) = 1$. This is the only swap that needs to occur so the answer is 1.

Function Description

Complete the function *rearrangeStudents* in the editor below.

rearrangeStudents has the following parameter:

- $int arrA[n]$: the heights of children in line A
- $int arrB[n]$: the heights of children in line B